

SaaS Business Performance Analysis Dashboard

Documentation

KPI Cards

1. Total Revenue

Purpose:

Displays the total revenue generated from all SaaS subscriptions.

Business Importance:

- Measures overall business performance.
- Indicates the company's revenue generation capability.
- Helps management monitor financial growth.

Business Use:

Executives use this KPI to evaluate overall company performance and compare revenue across different filters such as Region and Plan Category.

2. Total Customers

Purpose:

Displays the total number of unique customers.

Business Importance:

- Represents customer acquisition.
- Measures market reach.
- Indicates customer growth.

Business Use:

Used to evaluate customer acquisition strategies and business expansion.

3. Total Subscriptions

Purpose:

Displays the total number of subscriptions purchased.

Business Importance:

- Indicates product adoption.
- Measures recurring business.
- Helps understand subscription demand.

Business Use:

Used to evaluate subscription growth and recurring revenue potential

4. Average Payment

Purpose:

Displays the average payment received per subscription.

Business Importance:

- Measures average customer spending.
- Helps evaluate pricing strategy.
- Supports revenue optimization.

Business Use:

Used by management to understand customer purchasing behavior.

5. Average Discount

Purpose:

Displays the average discount percentage offered.

Business Importance:

- Evaluates promotional effectiveness.
- Helps monitor pricing strategy.
- Supports profitability analysis.

Business Use:

Used to balance customer acquisition with business profitability.

6. Total Tax

Purpose:

Displays the total tax collected.

Business Importance:

- Represents tax contribution.
- Helps monitor financial reporting.
- Supports compliance analysis.

Business Use:

Useful for financial reporting and taxation analysis.

Slicers

Plan Category

Purpose

Allows users to filter the dashboard by Premium and Standard subscription plans.

Business Value

- Compare performance between subscription plans.
- Analyze revenue and customer distribution.
- Evaluate plan adoption.

Region

Purpose

Filters all dashboard visuals based on customer region.

Business Value

- Compare regional business performance.
- Identify high-growth markets.
- Support expansion strategies.

Page 1 : Overview Section

1. Total Subscription Growth (Line Chart)

Purpose

Shows how subscriptions change over time.

Business Importance

- Identifies growth trends.
- Detects periods of high and low subscription activity.
- Measures business expansion.

Business Use

Management uses this chart to monitor subscription performance and identify seasonal demand.

2. Average Revenue Trend (Area Chart)

Purpose

Shows the average payment trend across subscription months.

Business Importance

- Evaluates revenue consistency.
- Identifies pricing fluctuations.
- Measures customer spending patterns.

Business Use

Helps monitor whether customer spending is increasing or decreasing over time.

3. Total Subscriptions by Subscription Year (Donut Chart)

Purpose

Displays subscription distribution across years.

Business Importance

- Compares yearly business growth.
- Shows yearly contribution.
- Identifies subscription concentration.

Business Use

Used to compare annual subscription performance.

4. Monthly Tax Collection (Area Chart)

Purpose

Displays tax collected each month.

Business Importance

- Indicates monthly business activity.
- Supports financial reporting.
- Reflects revenue fluctuations.

Business Use : Used to monitor taxation trends and business transactions.

Subscription Trends Section

1. Monthly Average Payment Trend (Line Chart)

Purpose

Shows changes in average payment over time.

Business Importance

- Measures customer spending behavior.
- Identifies seasonal pricing changes.
- Evaluates pricing effectiveness.

Business Use

Supports pricing optimization and revenue planning.

2. Revenue by Billing Cycle (Bar Chart)

Purpose

Compares revenue across Monthly, Quarterly, and Annual billing cycles.

Business Importance

- Identifies the most profitable billing model.
- Evaluates customer billing preferences.
- Supports pricing decisions.

Business Use

Helps management promote higher-value billing cycles.

3. Monthly Feature Usage (Column Chart)

Purpose

Displays feature usage across different years.

Business Importance

- Measures product engagement.
- Tracks feature adoption.
- Evaluates customer activity.

Business Use

Supports product improvement decisions.

4. Customer Satisfaction by Support Team (Bar Chart)

Purpose

Compares customer satisfaction across support teams.

Business Importance

- Evaluates service quality.
- Measures support performance.
- Identifies improvement opportunities.

Business Use

Helps improve customer experience.

Customer Segments Section

1. Customer by Segment (Bar Chart)

Purpose

Shows customer distribution across business segments.

Business Importance

- Identifies largest customer groups.
- Supports segmentation analysis.
- Helps prioritize target markets.

Business Use

Useful for targeted marketing strategies.

2. Average Company Revenue by Company Size (Column Chart)

Purpose

Compares average revenue across company sizes.

Business Importance

- Identifies high-value business customers.
- Evaluates enterprise contribution.
- Supports sales strategies.

Business Use

Helps prioritize enterprise customers.

3. Support Issues by Category (Bar Chart)

Purpose

Displays the number of support tickets across issue categories.

Business Importance

- Identifies frequent customer issues.
- Measures operational performance.
- Supports service improvement.

Business Use

Helps prioritize issue resolution and product improvements.

4. Feature Adoption by Customer Stage (Column Chart)

Purpose

Shows feature adoption across different customer lifecycle stages.

Business Importance

- Measures product engagement.
- Evaluates onboarding success.
- Identifies adoption opportunities.

Business Use : Supports customer success and product adoption strategies.

Business Insights Section

Executive Business Insights & Strategic Recommendations

Purpose

Provides executive-level interpretation of dashboard findings and converts analytical results into actionable business recommendations.

Business Importance

- Bridges the gap between data and decision-making.
- Highlights key business opportunities and risks.
- Supports strategic planning.

Business Use

Executives and managers can quickly understand the overall business performance and take informed actions based on the insights and recommendations provided.

Insights and Recommendations :

1. Premium Plan Performance

Business Insight:

Premium subscription plans demonstrate stronger customer adoption and contribute significantly to overall business performance, indicating higher perceived customer value.

Recommendation:

Enhance premium plan offerings and introduce targeted upselling campaigns to increase premium plan adoption.

2. Subscription Growth Trend

Business Insight:

Subscription growth varies across different periods, indicating fluctuations in customer demand and potential seasonal business patterns.

Recommendation:

Implement targeted marketing campaigns during slower periods to maintain consistent subscription growth.

3. Customer Segmentation

Business Insight:

Customer distribution differs across business segments, highlighting opportunities for segment-specific engagement and marketing strategies.

Recommendation:

Develop personalized communication and product offerings for each customer segment to improve customer retention.

4. Regional Performance

Business Insight:

Regional differences in customer adoption suggest that some markets offer greater expansion opportunities than others.

Recommendation:

Focus marketing investments and customer acquisition strategies on underperforming regions to increase market penetration.

5. Pricing & Revenue Strategy

Business Insight:

The current pricing and discount strategy supports customer acquisition while maintaining a sustainable revenue model.

Recommendation:

Continuously monitor pricing performance and optimize discount policies to maximize long-term profitability.

6. Customer Engagement

Business Insight:

Feature adoption and customer engagement vary across different customer stages, reflecting diverse product usage patterns.

Recommendation:

Strengthen onboarding and customer success programs to improve feature adoption and long-term engagement.

7. Support Performance

Business Insight:

Support requests are concentrated in specific issue categories, indicating opportunities to improve product usability and customer support processes.

Recommendation:

Prioritize resolving frequently reported issues and enhance self-service resources to improve the customer experience.

8. Business Growth**Business Insight:**

The combination of subscription growth, customer acquisition, and recurring revenue indicates a scalable SaaS business model with strong growth potential.

Recommendation:

Continue investing in customer retention, product innovation, and data-driven decision-making to achieve sustainable business growth.

Navigation Panel**Purpose**

Uses Power BI bookmarks to switch between:

- Overview
- Subscription Trends
- Customer Segments
- Business Insights

Business Importance

- Provides a clean, single-page dashboard experience.
- Improves usability.
- Enables quick access to different analytical views without navigating across multiple report pages.

Conclusion

The **SaaS Business Performance Analysis Dashboard** successfully transformed raw customer and operational data into meaningful business insights using Excel, SQL, Python, and Power BI. The project provided a comprehensive view of key business metrics, including revenue, subscriptions, customer growth, pricing strategy, and customer segmentation through an interactive and user-friendly dashboard.

By analyzing subscription trends, customer behavior, and regional performance, the dashboard identified valuable business opportunities and supported data-driven decision-making. The integration of executive business insights and strategic recommendations enables stakeholders to better understand business performance and take informed actions to improve customer engagement, optimize pricing strategies, and drive sustainable growth.

Overall, this project demonstrates the complete data analytics workflow, from data cleaning and transformation to visualization and business reporting, while showcasing practical skills in SQL, Python, Power BI, and business intelligence. The developed dashboard serves as an effective decision-support tool that helps organizations monitor performance, identify growth opportunities, and improve overall business efficiency.

Future Scope

- Integrate real-time SaaS data sources for live dashboard updates.
- Develop Machine Learning models for customer churn prediction.
- Implement Customer Lifetime Value (CLV) analysis for targeted marketing.
- Add forecasting models to predict subscription growth and revenue trends.
- Expand the dashboard with advanced analytics and automated reporting to support strategic business planning.